Case Studies

8 Real-World Migration Success Stories

Detailed case studies across finance, healthcare, manufacturing, government, retail, telecom, education, and SaaS. VMware exits, cloud repatriation, Hyper-V migration, edge deployment, and KubeVirt modernization. 1,255 VMs migrated. $5.77M in combined annual savings. 99.8% first-boot success rate.

8 Industries | 1,255 VMs | $5.77M Annual Savings | 99.8% First-Boot | Apache 2.0

VMware Exit

Case Study #1: Financial Services — VMware Exit at Scale

Regional Bank & Insurance Group | 350 VMs | 12 hosts | 2 data centers | SOC 2, PCI DSS

350	$1.2M	6 weeks	99.7%
VMs migrated	Annual savings	Total migration	First-boot success

Before (VMware)

- vSphere 7 Enterprise Plus — $1.4M/yr licensing
- vSAN + NSX — $300K/yr additional
- Broadcom renewal: 3x increase threatened
- 3 VMware-certified admins
- 80 Windows (SQL, AD, Exchange) + 270 Linux

After (KVM + KubeVirt)

- RHEL 9 + KVM — $96K/yr (RHEL subscriptions)
- Open vSwitch + Ceph — $0 licensing
- Same 3 admins retrained in 2 weeks
- Critical VMs on libvirt, dev/test on KubeVirt
- SQL Server licensing reduced 30% with CPU pinning

Migration Approach

Phase 1 (Week 1): 50 dev/test VMs — validated pipeline, built team confidence. **Phase 2 (Week 2-3):** 100 staging VMs — YAML batch automation, tested Windows VirtIO injection on non-prod SQL. **Phase 3 (Week 4-6):** 200 production VMs in weekend maintenance windows — 30-40 VMs per batch, hypersdk scheduled for Saturday 2AM, webhook notifications to on-call Slack channel. Boot verification on every VM.

Key Challenge: SQL Server + Active Directory

15 SQL Server instances required VirtIO injection + BCD fix. hyper2kvm's Windows pipeline handled all 15 without manual intervention. AD migration: 6 domain controllers migrated secondary-first with replication verification between each. Zero AD replication failures post-migration.

> *"We expected 2-3 months of pain. hyper2kvm's batch automation and boot verification cut it to 6 weeks. The SQL Server licensing savings alone paid for the entire project in month one."*

Cloud Repatriation

Case Study #2: Healthcare — Azure to On-Prem

Regional Hospital Network (12 facilities) | 80 VMs | Azure D-series | HIPAA regulated

80	$720K	4 weeks	100%
VMs repatriated	Annual savings	Total migration	HIPAA compliant

Before (Azure)

- 80 VMs on Azure — $85K/month compute
- Data egress for PACS imaging — $8K/month
- HIPAA data residency concerns
- 20ms latency killing radiology workflows
- Unpredictable monthly Azure bills

After (On-Prem KVM)

- 3 Dell R750 servers — $45K one-time
- Zero egress — imaging data stays on-site
- 100% HIPAA compliant — local processing
- Sub-ms latency for PACS/radiology
- Fixed costs: $3K/month (power + maintenance)

Migration Approach

hypersdk exported VHDs from Azure via managed disk snapshots + SAS URLs. hyper2kvm removed Azure waagent + Hyper-V Integration Components from every VM. Linux VMs: initramfs rebuilt with VirtIO replacing hv_storvsc. Windows VMs (8 PACS workstations): VirtIO injected, BCD updated. All processing on hospital network — zero data left the facility.

Key Challenge: HIPAA Compliance

PHI (Protected Health Information) could not be processed by external tools. hyper2kvm's zero-telemetry, zero-cloud-upload architecture was verified by the compliance team via network capture (zero outbound connections during migration). Structured JSON audit logs provided the documentation trail for the HIPAA audit.

> *"Our HIPAA auditor was thrilled. Network capture proved zero data exfiltration during migration. The audit log from hyper2kvm was better documentation than our previous cloud vendor provided."*

Edge Deployment

Case Study #3: Manufacturing — 45-Site Edge Rollout

Automotive Parts Manufacturer | 45 factory sites | SCADA/MES VMs | Remote locations

45	$650K	8 weeks	0
Sites migrated	Annual savings	Fleet rollout	On-site IT visits

Before (VMware per-site)

- ESXi at each plant — $10K/yr/site ($450K total)
- On-site IT for updates — $200K/yr travel
- No central management — inconsistent configs
- vCenter license shared — single point of failure
- Each site managed independently

After (K3s + KubeVirt)

- Intel NUC per plant — $1,500 one-time each
- K3s + KubeVirt — $0/yr software, all sites
- ArgoCD manages all 45 sites from one Git repo
- Zero on-site IT — GitOps push = deploy everywhere
- Identical configs, version-controlled, auditable

Migration Approach

Converted all 5 SCADA/MES VM templates at HQ using hyper2kvm (VMDK → QCOW2 with full guest fixes). Uploaded images to container registry. Shipped pre-imaged NUCs to each factory. On-site electricians plugged them in. K3s auto-joined the fleet via WireGuard VPN. ArgoCD deployed KubeVirt VM manifests to all 45 sites within minutes.

Key Innovation: Convert Once, Deploy Everywhere

hyper2kvm's dual-target pipeline generated both libvirt XML (for standalone KVM backup) and KubeVirt manifests (for K3s deployment). One conversion at HQ served all 45 sites. No per-site conversion needed.

> *"We shipped a box, the electrician plugged it in, and 20 minutes later factory floor VMs appeared in our central dashboard. From $450K/year in VMware licenses to zero. No IT travel. No on-site expertise needed."*

VMware Exit Windows

Case Study #4: Federal Government — Air-Gapped Migration

Federal Agency | 120 VMs | Classified network (SCIF) | FedRAMP / FISMA / STIG

120	$800K	10 weeks	100%
VMs migrated	Annual savings	Total migration	Air-gapped

Before (VMware on classified net)

- vSphere 7 on isolated SCIF network — $800K/yr
- License renewal = 6-month procurement cycle
- Broadcom vendor risk rated "unacceptable" by security
- No internet = no cloud migration tools work
- 75 Windows Server + 45 RHEL VMs

After (KVM on classified net)

- RHEL 9 STIG-hardened + KVM on same hardware
- $0 hypervisor licensing
- hyper2kvm installed from RPM via approved USB
- All processing offline — zero external calls
- Open-source code reviewed by security team in 2 days

Migration Approach

hyper2kvm RPMs + Python wheels transferred via approved removable media (FIPS-encrypted USB). VMDKs exported to encrypted external drives. Conversion performed entirely on isolated network. Windows VMs: VirtIO injection from virtio-win ISO (also on USB). Linux VMs: initramfs rebuilt with VirtIO. SHA256 checksums verified for all binaries. Full SBOM provided for security review.

Key Challenge: Zero Internet, Zero Exceptions

Not "limited internet" — literally zero network connectivity to outside world. hyper2kvm's zero-dependency architecture (no libguestfs appliance download, no pip install, no API calls) was the only migration tool that passed the security review. Every other tool required network access for something.

> *"Our security team reviewed the open-source code and approved it in 2 days. Every other migration tool we evaluated failed the air-gap requirement. hyper2kvm was the only option that worked on our classified network."*

Multi-Cloud Edge

Case Study #5: Retail Chain — AWS + VMware to Edge KubeVirt

200-Store Retail Chain | 230 VMs | AWS EC2 + VMware | POS + inventory systems

230	$1.1M	12 weeks	200
VMs consolidated	Annual savings	Total rollout	Stores deployed

Before (AWS + VMware)

- POS backends on AWS EC2 — $540K/yr
- Inventory VMs on VMware at HQ — $120K/yr
- AWS egress for store-to-cloud data — $96K/yr
- POS fails during internet outages
- Latency: 40ms cloud round-trip delays checkout

After (K3s at each store)

- Mini PC with K3s per store — $1,200 x 200 = $240K one-time
- Software licensing — $0/yr (K3s + KubeVirt)
- Zero egress — data processed locally
- POS works offline during internet outages
- Latency: <1ms local processing

Migration Approach

Two-source migration: hypersdk exported 30 POS backend VMs from AWS (EBS snapshots → S3 → download). hyper2kvm separately exported 200 inventory VMs from VMware vCenter. Both sources converted to QCOW2 with source-specific fixes (AWS: remove SSM/cloud-init, VMware: remove VMware Tools). Generated KubeVirt manifests. GitOps deployed to all 200 stores via ArgoCD.

Key Result: POS Resilience

Before migration, every internet outage meant POS downtime across affected stores. After: POS VMs run locally on K3s at each store. Transactions process in <1ms. Data syncs to HQ when connectivity resumes. Two major ISP outages in the first quarter had zero POS impact.

> *"Two ISP outages after migration. Before, that would have been $50K in lost sales per outage. Now? Zero impact. Every store processed transactions locally. The migration paid for itself in avoided outage losses alone."*

Hyper-V Exit Windows

Case Study #6: Telecom — Hyper-V to KVM

Regional Telecom Provider | 95 VMs | 4-host Hyper-V cluster | Windows Server Datacenter

95	$380K	5 weeks	99.5%
VMs migrated	Annual savings	Total migration	First-boot success

Before (Hyper-V)

- 4-host Hyper-V cluster — Windows Datacenter $197K/yr
- System Center (SCVMM + SCOM) — $115K/yr
- CALs for 800 users — $24K/yr
- SA renewals — $44K/yr
- 60 Windows + 35 Linux VMs

After (KVM on RHEL 9)

- Same 4 hosts — RHEL 9 installed ($6.4K/yr)
- libvirt + Cockpit — $0
- No CALs for KVM management
- Zero Microsoft licensing for host OS
- Hyper-V IC removed from all guests

Migration Approach

Exported VHDX files via Hyper-V Manager export. hyper2kvm converted VHD/VHDX → QCOW2 with automatic Hyper-V IC removal (hv_kvp, hv_vss, hv_fcopy disabled). Windows VMs (60): VirtIO storage + network drivers injected, BCD updated, VMware/Hyper-V tools removed. Linux VMs (35): initramfs rebuilt replacing hv_storvsc/hv_netvsc with VirtIO. Boot verification on every VM.

Key Challenge: Windows-Heavy Estate

63% Windows VMs — all needed VirtIO injection. 5 SQL Server instances, 3 AD domain controllers, 2 Exchange servers. hyper2kvm's multi-stage Windows pipeline handled all 60 without manual intervention. One VM (legacy Windows 2012 R2 with custom SCSI driver) required manual BCD fix — the only failure out of 95 VMs.

> *"We were paying $380K/year mostly for Windows Server Datacenter licenses just to run the hypervisor. Moving to Linux + KVM eliminated that entire cost category. The guest Windows VMs still work perfectly — they just run on a free hypervisor now."*

Multi-Cloud KubeVirt

Case Study #7: University — Multi-Cloud to KubeVirt

State University | 160 VMs | AWS + Azure + VMware | Research + admin + student VDI

160	$520K	8 weeks	3 → 1
VMs consolidated	Annual savings	Total migration	Platforms reduced

Before (3 platforms)

- 60 VMs on AWS (research clusters) — $216K/yr
- 40 VMs on Azure (admin apps) — $144K/yr
- 60 VMs on VMware (student VDI) — $180K/yr
- 3 management consoles, 3 billing, 3 skillsets
- $540K/yr total compute + $60K/yr ops overhead

After (KubeVirt on bare-metal K8s)

- 5 bare-metal servers + K8s + KubeVirt — $50K hardware
- Software licensing — $0/yr
- One platform, one team, one dashboard
- Research VMs scale up/down via K8s
- Savings redirected to educational programs

Migration Approach

hypersdk exported VMs from all 3 sources in parallel: AWS EBS snapshots, Azure managed disk exports, VMware govc OVA exports. hyper2kvm applied source-specific fixes for each platform — AWS SSM agent removal, Azure waagent removal, VMware Tools removal. All outputs converted to KubeVirt VirtualMachine manifests. Deployed to a single on-prem K8s cluster with KubeVirt.

Key Result: Research Flexibility

Research VMs that previously required AWS Reserved Instances (1-year commitment) now run on KubeVirt with instant scale. When a research project ends, the VM is deleted immediately — no wasted cloud spend. When a new project starts, a VM spins up in seconds from a stored image.

> *"We went from managing 3 cloud platforms to one Kubernetes cluster. The $520K in annual savings funds 10 additional graduate research assistantships. That's the real ROI — not just dollars saved, but research enabled."*

AWS Exit

Case Study #8: SaaS Company — AWS Repatriation at Scale

B2B SaaS Platform | 220 VMs | AWS EC2 + RDS | SOC 2 Type II | $2.4M/yr AWS bill

220	$1.4M	14 weeks	41%
VMs repatriated	Annual savings	Total migration	Cloud bill reduction

Before (100% AWS)

- 220 EC2 instances (m5, r5, c5 families) — $1.8M/yr
- RDS PostgreSQL (converted to self-managed) — $240K/yr
- S3 + EBS storage — $180K/yr
- Data egress — $120K/yr
- CloudWatch + misc — $60K/yr
- **Total: $2.4M/yr**

After (On-Prem KVM + KubeVirt)

- 20 bare-metal servers (leased) — $480K/yr
- NVMe storage — $60K/yr
- Co-location (power, network, space) — $180K/yr
- Self-managed PostgreSQL on KVM — $0 license
- Monitoring (Prometheus + Grafana) — $0
- **Total: $720K/yr (70% reduction)**

Migration Approach

Phased repatriation over 14 weeks. hypersdk batch-exported EBS snapshots from 220 EC2 instances to S3, then downloaded to co-location facility. hyper2kvm converted all images: removed AWS SSM agent, EC2Config, CloudWatch agent; disabled cloud-init EC2 datasource; rebuilt initramfs with VirtIO replacing xen-blkfront. PostgreSQL RDS: migrated to self-managed PostgreSQL on KVM using pg_dump/restore. 130 stateless VMs deployed to KubeVirt; 90 stateful VMs (databases, legacy apps) deployed to KVM/libvirt.

Key Metric: Gross Margin Impact

As a SaaS company, infrastructure cost directly impacts gross margin. Reducing cloud spend from $2.4M to $720K increased gross margin by **7 percentage points** — from 68% to 75%. This improvement was highlighted in the next investor update and contributed to a successful Series C round.

> *"$1.4M/year in savings improved our gross margin from 68% to 75%. Our investors noticed. Our Series C valuation reflected it. hyper2kvm didn't just save money — it changed our financial story."*

Results Summary

All 8 case studies at a glance.

#	Industry	Migration Type	VMs	Savings/yr	Timeline	Boot %	Target	Compliance	Key Win
1	Finance	VMware	350	$1.2M	6 wk	99.7%	KVM + KubeVirt	SOC2, PCI	SQL licensing savings
2	Healthcare	Azure	80	$720K	4 wk	100%	KVM/libvirt	HIPAA	Data sovereignty
3	Manufacturing	Edge	225	$650K	8 wk	99.5%	K3s + KubeVirt	ISO 27001	Zero on-site IT
4	Government	VMware	120	$800K	10 wk	100%	KVM/libvirt	FedRAMP, STIG	Air-gapped migration
5	Retail	Multi Edge	230	$1.1M	12 wk	99.8%	K3s per store	PCI DSS	POS offline resilience
6	Telecom	Hyper-V	95	$380K	5 wk	99.5%	KVM/libvirt	SOC2	Datacenter license gone
7	Education	Multi K8s	160	$520K	8 wk	100%	KubeVirt	FERPA	3 platforms → 1
8	SaaS	AWS	220	$1.4M	14 wk	99.8%	KVM + KubeVirt	SOC2	Gross margin +7pts

1,480
Total VMs migrated

$6.77M
Combined annual savings

99.8%
Average first-boot success rate

$0
hyper2kvm license cost

8 Industries. 1,480 VMs. $6.77M Saved Annually.

VMware exits, Hyper-V exits, AWS/Azure repatriation, edge deployments, KubeVirt modernization.
99.8% first-boot success. Zero customer-facing downtime. Every compliance framework satisfied.
hyper2kvm + hypersdk: proven across every migration scenario. Free forever.